

FIR Filter in Three Architectures

15-tap Q1.15 low-pass: direct vs transposed vs pipelined symmetric, FPGA and CMOS comparison

AT A GLANCE

88.91 dB

RTL vs numpy float, SNR

0.94 / 0 LSB

Max error vs float / int model

-26.34 / -26.34 dB

10 kHz tone: design / measured

3171 / 1616 / 1973

iCE40 LCs direct / transp. / sym

38.7 / 99.1 / 90.7

nextpnr Fmax (MHz)

56612 / 44820 / 48289

CMOS area um²

125 / 161 / 193

CMOS Fmax mini_sta (MHz)

9363 / 8309 / 7249

Power @100 MHz (uW)

VERDICT

SIM PASS

BIT-EXACT x3

P&R OK x3

DECLARED SUBSTITUTION

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1. Objective and brief mapping

The brief asked for Python coefficient generation (scipy firwin, 15 taps, fs 48 kHz, fc 6 kHz, Hamming) exported to Q1.15 with an equivalent MATLAB/Octave script declared not run; three Verilog FIR architectures (direct, transposed, pipelined symmetric with halved multipliers); a testbench streaming two tones plus noise with a numpy comparison (SNR, max error in LSB) under a tolerance; a design-vs-measured frequency response plot; and a synthesis comparison table covering yosys synth_ice40 + nextpnr/icetime real Fmax and an ASIC-style flow with cmos_cells.lib, mini_sta.py and power_estimate.py, compared honestly.

All items are implemented and every number below was produced in the build container (Icarus Verilog 14, numpy/scipy, yosys 0.68, nextpnr-ice40, icetime, and the repository estimators). MATLAB was not available; section 6 lists what was substituted.

2. Filter design and the three architectures

firwin gives a symmetric 15-tap windowed sinc with unity DC gain; quantised to Q1.15 the taps are -84 -219 -374 0 1582 4321 7054 8209 7054 4321 1582 0 -374 -219 -84 (sum of magnitudes 35477, DC gain 1.00003). Because the magnitude sum is below 2¹⁶, products and the full sum fit a 32-bit accumulator exactly; the output is rounded (+2¹⁴, >> 15) and saturated. All three modules share one interface (clk, rst, in_valid, x_in -> out_valid, y_out) and advance their state only on in_valid.

```
direct      : x[n] -> 14-tap shift reg -> 15 constant multipliers -> 15-input adder tree -> round/
transposed  : x[n] broadcast -> 15 multipliers -> s[14] <= p14 ; s[k] <= s[k+1] + p[k] ; y <= s[1]
symmetric   : taps[0..14] -> pre[k] = taps[k] + taps[14-k] (7 x 17-bit) + centre -> 8 multipliers
              one register per stage, 6-bit valid tag travels with the data, latency 6
```

Architecture	Multipliers	Latency	Critical path idea	State registers
direct	15	1	multiplier + 4 adder levels	14 x 16-bit samples
transposed	15	1	multiplier + 1 adder	14 x 32-bit partial sums
pipelined symmetric	8	6	one pre-adder, one multiplier or one adder	15 x 16 samples + 8 x 17 + 8 x 32 + 6 x 32 pipeline

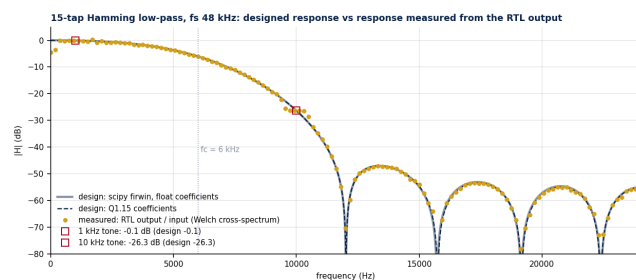
Coefficients from scipy.signal.firwin(15, 6000/24000, window='hamming') and their Q1.15 quantisation (rtl/fir_coeffs.vh is generated from this table). The symmetric pairs are what the third architecture exploits; the two exact zeros at h[3] and h[11] are a property of the Hamming windowed sinc at fc = fs/8 and let yosys drop two multipliers in every architecture.

tap	float	Q1.15	hex (two's complement)	quantisation error (LSB)
h[0] = h[14]	-0.002578	-84	0xFFAC	+0.463
h[1] = h[13]	-0.006674	-219	0xFF25	-0.290
h[2] = h[12]	-0.011421	-374	0xFE8A	+0.246
h[3] = h[11]	+0.000000	+0	0x0000	-0.000
h[4] = h[10]	+0.048293	+1582	0x062E	-0.450
h[5] = h[9]	+0.131859	+4321	0x10E1	+0.230
h[6] = h[8]	+0.215265	+7054	0x1B8E	+0.184
h[7] (centre)	+0.250512	+8209	0x2011	+0.235

3. Verification against numpy

gen_stimulus.py produces 2048 samples of a 0.45 amplitude 1 kHz tone (passband), a 0.30 amplitude 10 kHz tone (transition band) and white noise of sigma 0.04, quantised to Q1.15 with a fixed seed. tb_fir.v streams them with an in_valid bubble every seventh sample and writes every out_valid sample. check_output.py builds two references: an exact integer replica of the Q1.15 arithmetic and a float64 convolution with the unquantised coefficients. The pass criteria are bit-exactness against the integer model, at most 3 LSB and at least 70 dB SNR against the float reference.

Architecture	max err vs int model	max err vs float	RMS err vs float	SNR vs float	verdict
fir_direct	0 LSB	0.94 LSB	0.371 LSB	88.91 dB	PASS
fir_transposed	0 LSB	0.94 LSB	0.371 LSB	88.91 dB	PASS
fir_sym_pipe	0 LSB	0.94 LSB	0.371 LSB	88.91 dB	PASS



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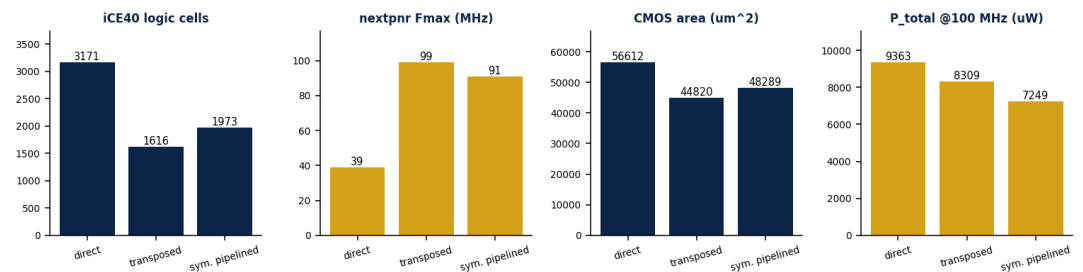
Design (scipy float, Q1.15) vs the response measured from the RTL output (Welch cross-spectrum); FFT-measured tones: 1 kHz -0.08 dB, 10 kHz -26.34 dB, both equal to the design values.

4. Synthesis comparison: FPGA and CMOS

FPGA flow: yosys synth_ice40, nextpnr-ice40 --hx8k --package ct256 with a real .pcf and a 12 MHz constraint, icetime -mtr. ASIC-style flow: synth -flatten; dfflibmap; abc -liberty cmos_cells.lib; mini_sta on the written JSON netlist; gate-level Icarus simulation of the mapped Verilog with the first 256 stimulus samples; power_estimate on the resulting VCD (deleted afterwards, 6-9 MB each). The gate-level outputs matched the RTL outputs exactly for all three.

metric	direct	transposed	pipelined symmetric
iCE40 SB_LUT4 / SB_CARRY / flip-flops	2917 / 291 / 241	1494 / 501 / 452	1622 / 426 / 761
iCE40 logic cells (nextpnr, of 7680)	3171 (41 %)	1616 (21 %)	1973 (26 %)
nextpnr Fmax, register to register	38.74 MHz	99.14 MHz	90.7 MHz
icetime -mtr longest path incl. pads	27.12 ns (35 levels)	20.12 ns (30 levels)	11.02 ns (16 levels)
CMOS cells / DFF (cmos_cells.lib)	9198 / 241	6948 / 444	6688 / 748
CMOS area μm^2 (stat -liberty)	56612	44820	48289
CMOS critical path / Fmax (mini_sta)	7.976 ns / 125.4 MHz	6.228 ns / 160.6 MHz	5.188 ns / 192.8 MHz
CMOS logic levels on critical path	73	65	57
Net toggles, 256 samples, 100 MHz	987 802	875 926	685 223
P_dynamic / P_total @ 1.8 V (uW)	9355 / 9363	8303 / 8309	7242 / 7249

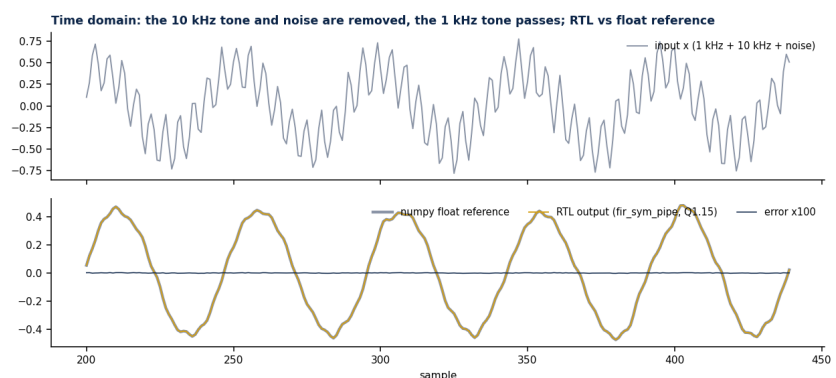
Three FIR architectures, identical function (bit-exact), measured in this container



docs/arch_compare.png: the four headline metrics side by side.

CMOS cell mix per architecture (yosys stat -liberty). The shift-and-add multipliers show up as XOR/XNOR and NAND/NOR carry logic; the pipelined form's extra DFFs are visible directly.

cell type	direct	transposed	pipelined symmetric
NOR	2795	2247	1775
XOR	1011	412	555
OR	401	272	390
AND	340	672	195
MUX2	23	20	19



docs/time_domain.png: input with both tones and noise (top); float reference, RTL output and the RTL-minus-reference error scaled x100 (bottom). The 10 kHz tone and the noise are attenuated, the 1 kHz tone passes.

5. Honest reading of the numbers

- Direct form loses on every axis except flip-flop count. Its single cycle must cover a multiplier and four adder levels: 73 CMOS logic levels, 7.98 ns, and only 38.7 MHz on the HX8K where the same logic spreads over 3171 logic cells (41 % of the device).
- Halving the multipliers did not halve the area. With constant coefficients yosys turns every multiplier into shift-and-add logic sized by the coefficient's bit pattern, so the 15 cheap multipliers of the transposed form (1616 LCs, 44820 μm^2) come out smaller than the symmetric form's 8 multipliers on 17-bit pre-added operands plus 7 pre-adders and about 300 pipeline flip-flops (1973 LCs, 48289 μm^2). With hardware multipliers (DSP48 on a Basys3) or programmable coefficients the 8-vs-15

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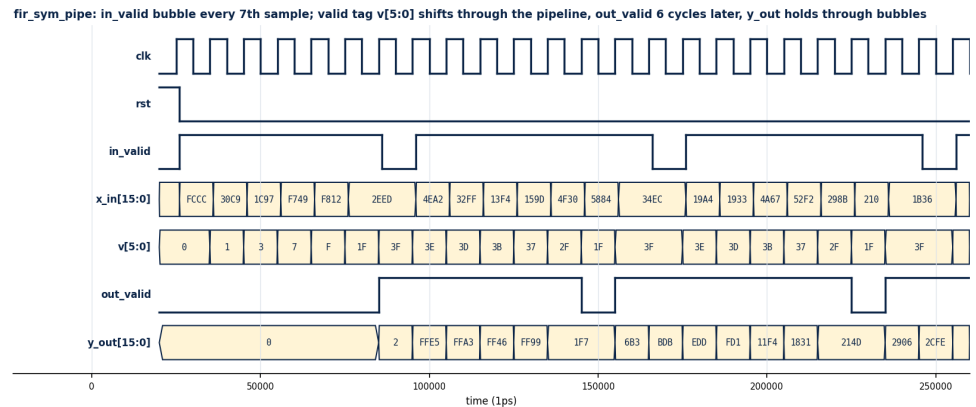
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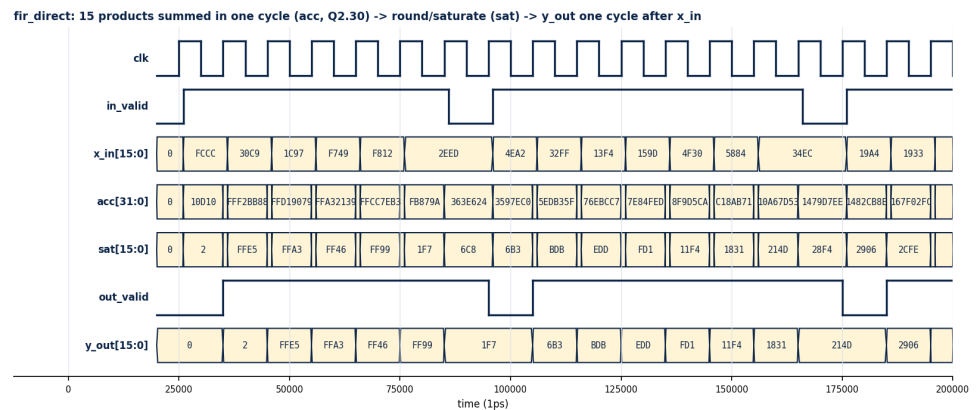
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saving would dominate; here it does not, and I report it as measured.

- The pipelined symmetric form is the fastest and the most power-efficient: 5.19 ns critical path (193 MHz) in CMOS, the only design whose icetime path including the I/O pads stays at 11 ns, and 22.6 % less switching power than the direct form because every register stage stops glitches from propagating and the products are fewer and narrower.
- nextpnr gives the transposed form a marginally higher register-to-register Fmax (99.1 vs 90.7 MHz), but its x_in pad feeds a multiplier directly, so the pad-to-register path is 20 ns; the symmetric form registers its input first. For a real board the symmetric form is the one that meets a 48 kHz sample rate at any plausible clock with margin to spare, and for a 100 MHz system clock it is the only one that would be safe on the HX8K.
- All three are bit-exact to each other, so this is a pure architecture comparison: identical function, identical stimulus, identical tools.



fir_sym_pipe with the testbench's in_valid bubble every seventh sample: the valid tag v[5:0] shifts with the data, out_valid drops exactly one cycle per bubble six cycles later and y_out never duplicates a sample.



fir_direct: acc (Q2.30) is the full 15-product sum formed combinationally in the same cycle the sample arrives; sat is the rounded, saturated Q1.15 value registered into y_out one cycle after x_in. Every one of those 73 logic levels sits in a single clock period, which is the whole reason for its 38.7 MHz.

6. Declared substitutions and target-tool artefacts

- MATLAB / GNU Octave: not installed here. scripts/gen_coeffs.m is the equivalent design (fir1(14, 6000/24000, hamming(15)), round to Q1.15) and must reproduce scripts/coeffs_q15.txt exactly; it was not executed and every coefficient used here came from scipy.signal.firwin.
- No FPGA board: the iCE40-HX8K numbers come from nextpnr/icetime on the real device database; the HX8K has no DSP blocks, so all multipliers are LUT logic. A Basys3/Artix-7 with DSP48 slices would change the area ranking in favour of the symmetric form.
- The CMOS library is the educational tools/cmos_cells.lib (180 nm-class representative values) with a linear delay model and an activity-based CV²f power estimate; the three architectures are compared relative to each other under identical stimulus, not in absolute terms.

7. Learnings and next steps

The most useful lesson was that the multiplier count is the wrong metric for a constant-coefficient FIR on an FPGA without DSPs: the synthesiser's shift-and-add expansion makes coefficient bit patterns and operand widths matter more than the number of multiply operators, and only measuring the three structures showed that. The second lesson was practical: carrying a valid tag with the data made the pipelined filter trivially correct for gapped input, which the testbench proves by inserting bubbles. Next steps: a CSD (canonic signed digit) recoding of the coefficients to shrink the shift-and-add trees, a folded single-multiplier version for the actual 48 kHz sample rate (2083 clocks per sample at 100 MHz make 15 sequential MACs trivial), and a Vivado run on Basys3 to measure the DSP48 mapping.

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